Data processing

1 Introduction

This supplementary document provides a detailed description of the stages of the data processing pipeline. The stages are: 1) ground truth generation; and 2) prediction post-processing.

2 Ground truth generation

The raw dataset recorded as bag files is processed to generate ground truth files that match the following requirements:

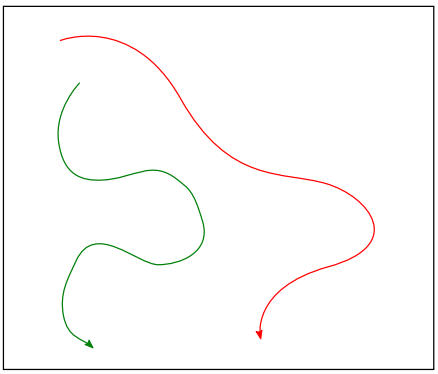
- timestamp are numbered sequentially,
- no timestamps when people are not visible by any sensor,
- no timestamps when the motion-capture loses track of any object (which may happens because of occlusions when people/platform are very close),
- different ID each time a person is not visible for a while (6 consecutive frames) to support trackers without re-identification capability,
- detection as boxes of width 0.5 m centered on the coordinates of the detection to support metrics based on intersections over unions (IoU).

Figure 1 shows the operations required to achieve this result.

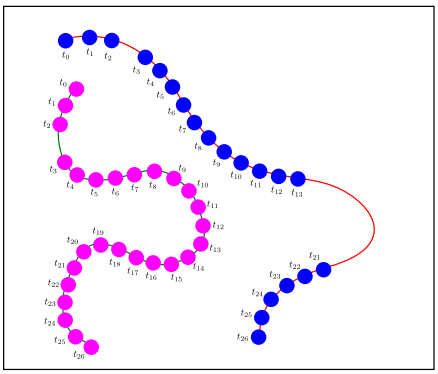
2.1 Representing raw data

Figure 1a shows the projection on the floor of the path followed by the people involved in an experiment. In this particular example, there are two paths, one in green and one in red. The two paths end at the tip of the line with the arrow and start from the other tip.

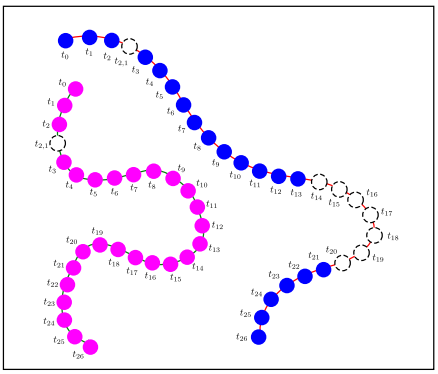
The sequences of points that overlap the paths in Figure 1b, represent the detections of people made by the motion capture system (Optitrack) over time. Each detection is associated with its capture timestamp, $t \in \{t_0, \cdots, t_{26}\}$ where $\forall i \in [0, 25] t_i < t_{i+1}$. Since the Optitrack captures the position of all the targets at each reading, the detections of the two people share the same set of timestamps, i.e. both start with t_0 and end with t_{26}. There is an exception for timestamps between t_{14} and t_{20}: for this time interval, there are no detections for the red path. This can happen because, in some circumstances, the Optitrack can lose one or more targets. We also highlight that between t_2 and t_3 the Optitrack skipped a capture, and one detection is missing for both paths.



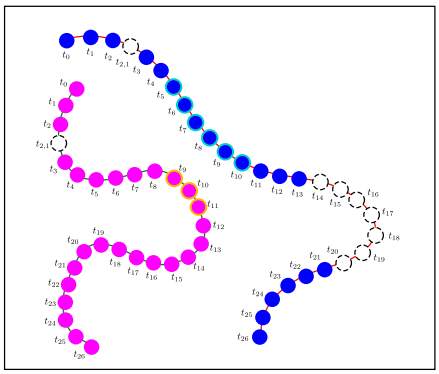
(a) Real traces



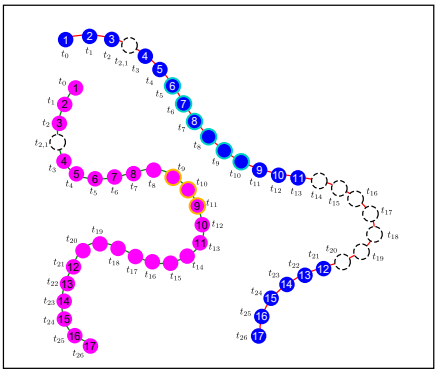
(b) Optitrack detections



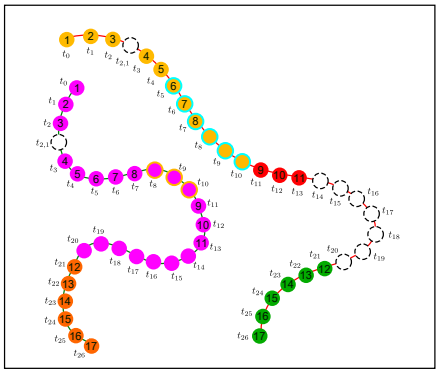
(c) Filling missing timestamps



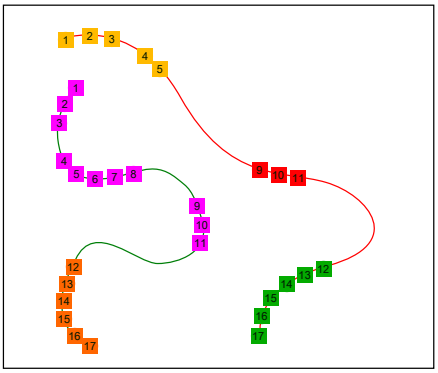
(d) Marking out of FOV targets



(e) Associating sequential numbers



(f) Splitting traces



(g) Saving ground truth

Figure 1: Process to generate the ground truth

2.2 Filling the gaps

Figure 1c shows the missing detections with dotted circles, each of which is associated with a timestamp. Those missing exclusively from the red path because the Optitrack has temporarily lost the corresponding target are associated with the timestamp of the captures occurring between t_{14} and t_{20}, those missing because the tracker has skipped a capture are associated with $t_{2,1} = t_2 + 1/r$ where r is the capture frequency of the Optitrack.

2.3 Excluding out of FOV targets

For each detection, we compute whether it was in the FOV (Field Of View) of at least one wheelchair sensor. This computation is possible because the Optitrack tracks the position and orientation of the wheelchair and the position of the sensors with respect to the wheelchair is known. Detections that are not in the FOV of any sensor are marked with a turquoise circle in Figure 1d.

2.4 Numbering the timestamps

Since the benchmark toolkit requires timestamps to be replaced with sequential numbers, we assign a number to each timestamp. We start from 1 and consider in the numbering only the timestamps that satisfy these conditions:

- has at least one person in the FOV of one sensor of the wheelchair,
- in that timestamp the Optitrack has detected all the targets: people and wheelchair.

In our example, the first condition excludes from the numbering t_9, t_{10} and t_{11}; while the second condition excludes the timestamp $t_{2,1}$ and the timestamps between t_{14} and t_{20}. The numbers associated with the timestamps are shown if Figure 1f inside the detections occurred during each timestamp. The association between timestamps and numbers is stored in a dedicated file because it is used in the processing described in Section 3. This file also stores the timestamps that are not associated with a number.

2.5 Splitting the original tracks

As the evaluation of people trackers with re-identification features is not taken into account, we assign different IDs to detections relating to the same person if they are out of the field of view (FOV) for more than six consecutive detections, or if more than six consecutive timestamps are not numbered in previous steps (or a combination of the two). This operation is shown in Figure 1e and changes the ID of the detections associated with the red path twice: once at t_{11}, after six consecutive detections that were out of FOV, and once at t_{21} after seven consecutive timestamps not associated with a number. The ID of the detections associated to the green path changes only once, after seven consecutive timestamps not associated with a number. In Figure 1g detections with the same ID share the same color.

2.6 Storing the ground truth

The final step stores in the ground truth the detections that satisfy the following conditions:

- the detection falls in the FOV of at least one of the sensors of the wheelchair,
- the detections has been done in a timestamp associated to a number.

So far, the detection were dimensionless points. To allow the computation of those metrics that are based on the concept of intersection over union (IoU), we transform them into boxes of width 0.5 m centered on the coordinates of the detection. For each detection, the following information is stored:

- number associated to the timestamp,
- x and y coordinates of the bottom left vertex of the square associated to the detection,
- width of the square associated to the detection.

3 Prediction post-processing

The output of the people tracker is expected to be a text file in semicolon-separated values format with header. Each row of the file provides when and where a person has been detected and to which track has been associated. In detail, these are the columns of the file:

- *id*: ID of the track,
- x: x position in meters of the detection,
- y: y position in meters of the detection,
- *timestamp*: when the detection took place.

The file with the predictions, before being evaluated against the ground truth, is processed as described in the following and shown in Figure 2. This processing is necessary for the following reasons:

- We want to be able to evaluate people trackers that work at different frequencies,
- Given that in the ground-truth dataset, we are breaking the tracks if more than six consecutive detections are missing, we do the same for the predicted tracks.

For clarity, in this explanation, we kept only one of the two tracks present in Figure 1.

3.1 Representing predicted data

Figure 2a shows the predicted detection with their timestamp. Timestamps are represented with the letter t' and numbered such that for t'_i, t_i is the closest timestamp in the ground truth. Detections share the same color if the people tracker associates them with the same track. Note that in this example, the people tracker is working half of the frequency of the Optitrack. Note also that there are predicted detections also for timestamps between t_{14}, t_{20}. This is because, even if the Optitrack missed this target, sensor data is present, and the people tracker generated detections also in this time window.

3.2 Associating predicted and ground truth timestamps

The timestamps that appear in the predictions are associated with the closest timestamps that appear in the ground truth (exploiting the information stored in the file generated by the process described in Section 2.4). Because of the definition of t'_i, it always holds that t'_i gets associated to t_i. In Figure 2b we circle with red lines, predictions, and Optitrack scans that took place in timestamps that have been associated.

3.3 Interpolating the predictions

In Figure 2c interpolated detections (represented with dotted circles) are generated for a predicted track s at each ground-truth timestamp t_i for which it holds that:

- the track has no prediction associated to t_i,
- it exists t_j and t_v such that $j < i < v$ and s has predicted detection in t'_j and t'_v.

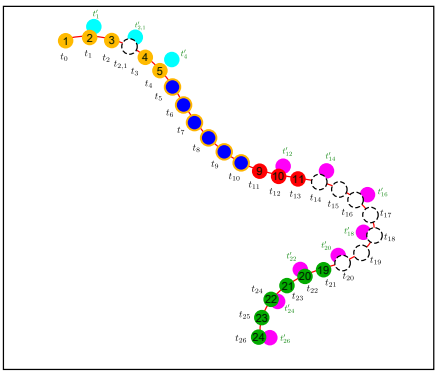
The interpolated detection timestamp is set to t_i and is associated with the ground truth detection that shares the same timestamp, as shown in Figure 2d.

3.4 Splitting the predicted tracks

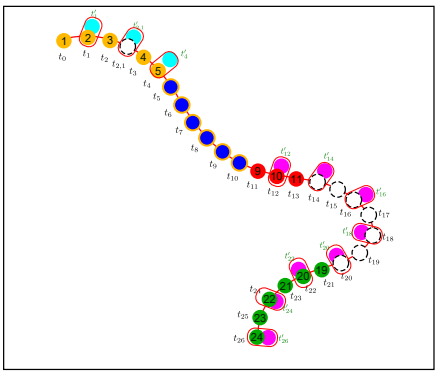
The predicted tracks are split in the same way as the ground-truth tracks. As can be seen in Figure 2e, the pink trace changes color because more than six timestamps were recorded during which the Optitrack did not work properly. This prevents a people tracker prediction from being penalized because a track has been broken in the ground truth but not in the prediction.

3.5 Storing the prediction

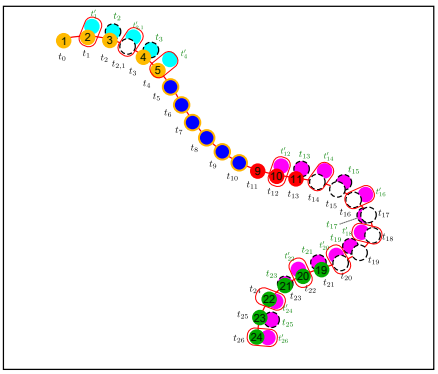
As shown in Figure 2f, the sequential numbers used to number the timestamps in the ground truth and shown on the detections are inherited by the predicted detections. Finally, Figure 1g shows the detections that are saved for evaluation. Note that they are saved only if their timestamp is associated with one in the ground truth for which all the targets have been tracked by the Optitrack. The output is stored in the format requested by the benchmark toolkit.



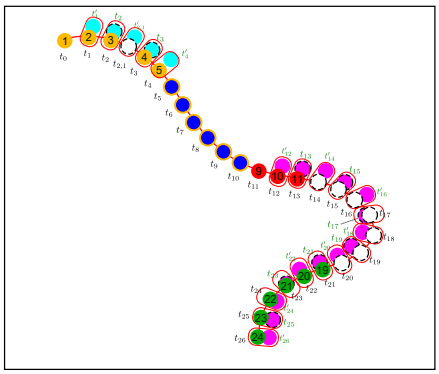
(a) Ground truth and predicted detections



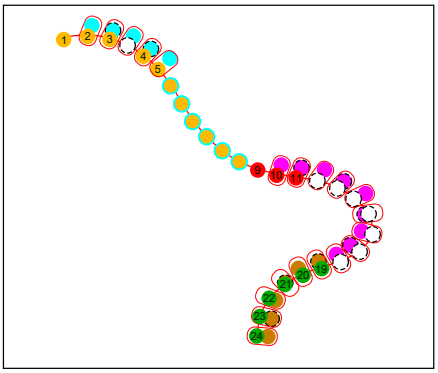
(b) Associating ground truth and predicted detections



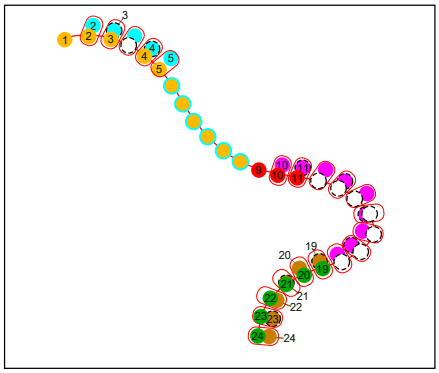
(c) Interpolating the predictions



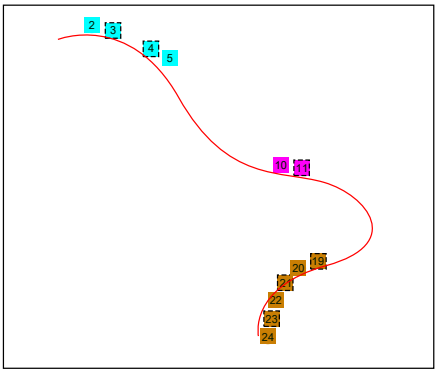
(d) Assocaiting interpolated detections



(e) Splitting the predicted tracks



(f) Associating sequential numbers



(g) Saving predictions

Figure 2: Prediction pre-processing